

Assignment

LED

1. Consider that last digit of your Student ID is “x”. Connect two LEDs of different color at digital pin “x” and “9 - x”. Write a code so that they blink in the following direction indefinitely –

Time	LED 1 (at Dpin x)	LED 2 (at Dpin 9-x)
0 – 1 s	ON	OFF
1 – 2 s	OFF	OFF
2 – 3 s	OFF	ON
3 – 4 s	OFF	OFF
4 – 5 s	ON	ON
5 – 6 s	OFF	OFF

2. Consider 2 LEDs of different colors are connected to two digital pins. Take two integer numbers (A & B) as user inputs. Now depending on the number following situation will occur –

Input Number	LED 1	LED 2
A is Divisible by B	ON	OFF
Otherwise	OFF	ON

3. Design a priority encoder ($D_3 > D_2 > D_1 > D_0$). Take a 4-digit user input (valid binary combination). Then show the output using two LEDs. Print the word “INVALID” if any user gives 0000 as Input.

User Input	LED 1	LED 0
1111	ON	ON
0010	OFF	ON
0101	ON	OFF
0001	OFF	OFF

4. Take two integer inputs from user. After that take a String input from User. Depending on the string input, perform the relevant arithmetic operation using the two input numbers. If a user gives wrong string name for arithmetic operation, it would be indicated by turning on a LED. For valid string inputs LED would be off.

String	Operation
Add or +	$A + B$
Subtract or -	$A - B$
Multiply or *	$A * B$
Division or /	A / B
Modulus or %	$A \% B$
Other	LED would be ON

5. Consider we would take an integer input (n) from user. Now write a custom function so that LED would blink $n!$ (Factorial) times. However, if $n! > 50$, it would discard the input and prompt the user to give another input.

Input Value, $n = 3$	(LED Would Blink $3! = 6$ times)
Input Value, $n = 4$	(LED Would blink $4! = 24$ times)
Input Value, $n = 5$	(as $5! = 120 > 50$, it would prompt user to give another input instead of blinking)

6. Take a 3 Digit Integer Input from a User ($A = A_3A_2A_1$). Consider we have two LEDs connected with Arduino board. Now depending on the user input following events would take place –

Situation	LED 1	LED 2
$A_3 > A_1$	Blink A_2 Times	OFF
$A_3 < A_1$	OFF	Blink A_2 Times
$A_3 = A_1$	OFF	OFF

7. Write a custom function that would take 3 inputs from user. First input would indicate no. of blinks, second input would indicate ON time in second and the third input would mean the duty cycle in percentage.

Custom Function Format: blinkLED (a, b, c)

Example – blinkLED (3, 2, 60)

a = 3; // LED would blink 3 times

b = 2; // LED would be ON for 2 second

c = 40; // 40% ON time (Duty Cycle), so LED would be OFF for 60% (= 3s)